



WORK SHEET - 31

QUADRATIC EQUATION & LOGARITHM

Q.1 (C)

$$x^2 - 10x + 21 = 0$$

Q.2 (B)

$$\text{roots} \rightarrow \underset{\alpha}{2\omega^2}, \quad \underset{\beta}{3+4\omega}, \quad \underset{\gamma}{3+4\omega^2}, \quad \underset{\delta}{5-\omega-\omega^2}$$

$$\delta = 5 - (\omega + \omega^2) = 5 - (-1) = 6$$

If $\alpha = 2\omega^2$ is a root then 2ω has to be a root too.

total $\rightarrow$ min. 5 roots, hence min. degree $\rightarrow 5$

Q.3 (B)

$$(x-41)^{49} + (x-49)^{41} + (x-2013)^{2013} = 0$$

$$f(x) = (x-41)^{49} + (x-49)^{41} + (x-2013)^{2013}$$

$$f'(x) = 49(x-41)^{48} + 41(x-49)^{40} + 2013(x-2013)^{2012} > 0$$

Hence, $f(x)$ will cut x-axis only once.

Q.4 (C)

$$x \notin I \quad \text{and} \quad [x] > 1$$

$$\Rightarrow x \in (2, 3) \text{ only option satisfy}$$

Q.5 (C)

$$2x^3 + ax + b \neq 2x^3 + cx + d$$

$$(a-c)x \neq d-b$$

$$\text{If } a-c=0 \text{ and } d-b \neq 0$$

$$\therefore \text{Max. } (a-c)^2 + (b-d) = 0 + 5 \text{ Ans.}$$

Q.6 (C)

$$f(x) = x^{2n+1} - (2n+1)x + a$$

$$f'(x) = (2n+1)(x^{2n} - 1)$$

$$x = -1 \rightarrow \text{point of local maxima}$$

$$x = -1 \rightarrow \text{point of local minima}$$

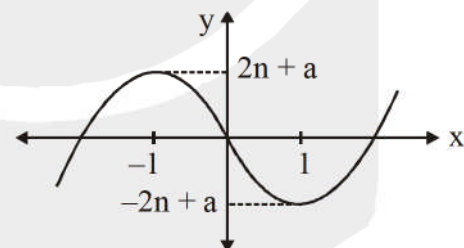
$$\text{Graph of } y = f(x) \text{ when } a = 0$$

$$f(x) \text{ is decreasing on } [-1, 1].$$

$$\text{Thus, } f(x) \text{ will have at most one root one } [-1, 1]$$

$$\text{that too will happen if } f(-1) \geq 0 \text{ and } f(1) \leq 0$$

$$\text{i.e. for } a \in [-2n, 2n].$$



Q.7 (B)

$$f(x) = 3x^2 - 2(a_1 + a_2 + a_3)x + a_1a_2 + a_2a_3 + a_3a_1 \geq 0$$

$$\therefore D \leq 0$$

$$4(a_1 + a_2 + a_3)^2 - 4.3.(a_1a_2 + a_2a_3 + a_3a_1) \leq 0$$

$$a_1^2 + a_2^2 + a_3^2 - a_1a_2 - a_2a_3 - a_3a_1 \leq 0$$

$$\frac{1}{2}[(a_1 - a_2)^2 + (a_2 - a_3)^2 + (a_3 - a_1)^2] \leq 0$$

$$\therefore a_1 = a_2 = a_3$$



Q.8 (A)

$$(p-5)x^2 - 2px + (p-4) = 0 \quad \text{or} \quad x^2 - \left(\frac{2p}{p-5}\right)x + \left(\frac{p-4}{p-5}\right) = 0$$

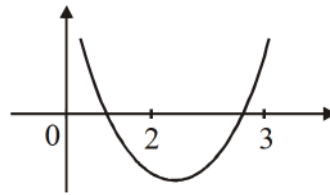
$$f(x) = x^2 - \left(\frac{2p}{p-5}\right)x + \left(\frac{p-4}{p-5}\right)$$

$$f(0) > 0, f(2) < 0, f(3) > 0$$

$$f(0) > 0 \Rightarrow \frac{p-4}{p-5} > 0 \quad \dots\dots(i)$$

$$f(2) < 0 \Rightarrow \frac{p-24}{p-5} < 0 \quad \dots\dots(ii)$$

$$f(3) > 0 \Rightarrow \frac{4p-49}{p-5} > 0 \quad \dots\dots(iii)$$



Intersection of (i), (ii) and (iii) gives $p \in \left(\frac{49}{4}, 24\right)$ **Ans.**

Q.9 (D)

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$$

$$(a+b+c)x^2 - (b+2c)x + c = 0$$

$$\left(1 + \frac{b}{a} + \frac{c}{a}\right)x^2 - \left(\frac{b}{a} + \frac{2c}{a}\right)x + \frac{c}{a} = 0$$

$$(1 - \alpha - \beta + \alpha\beta)x^2 + (\alpha + \beta - 2\alpha\beta)x + \alpha\beta = 0$$

$$x^2 + \left(\frac{\alpha + \beta - \alpha\beta - \alpha\beta}{1 - \alpha - \beta + \alpha\beta}\right)x + \left(\frac{\alpha\beta}{1 - \alpha - \beta + \alpha\beta}\right) = 0$$

$$x^2 + \left(\frac{\alpha(1-\beta) + \beta(1-\alpha)}{(1-\alpha)(1-\beta)}\right)x + \left(\frac{\alpha\beta}{(1-\alpha)(1-\beta)}\right) = 0$$

$$x^2 - \left[-\frac{\alpha}{1-\alpha} - \frac{\beta}{1-\beta}\right]x + \left(-\frac{\alpha}{1-\alpha}\right)\left(-\frac{\beta}{1-\beta}\right) = 0$$

$$x^2 - \left(\frac{\alpha}{\alpha-1} + \frac{\beta}{\beta-1}\right)x + \left(\frac{\alpha}{\alpha-1}\right)\left(\frac{\beta}{\beta-1}\right) = 0$$

$$\Rightarrow \text{root } \frac{\alpha}{\alpha-1} \text{ and } \frac{\beta}{\beta-1}$$

Alternative :

$$(a+b+c)x^2 - (b+2c)x + c = 0$$

$$ax^2 + b(x^2 - x) + c(x^2 - 2x + 1) = 0$$

$$a\left(\frac{x}{x-1}\right)^2 + b\left(\frac{x}{x-1}\right) + c = 0$$

Put $t = \frac{x}{x-1}$

$$at^2 + bt + c = 0 \Rightarrow t = \alpha, t = \beta$$

$$\Rightarrow \frac{x}{x-1} = \alpha, \frac{x}{x-1} = \beta$$

$$x = \frac{\alpha}{\alpha-1}, x = \frac{\beta}{\beta-1} \quad \text{Ans.}$$

Q.10 (B)

$$T_{r+1} = {}^{15}C_r \cdot (2)^r \cdot (x)^{30-3r}$$

$$\therefore \text{Ratio} = \frac{{}^{15}C_5 \cdot (2)^5}{{}^{15}C_{10} \cdot (2)^{10}} = \frac{1}{2^5} = 1 : 32. \text{ Ans.}$$

Q.11 (B)

$$|\text{adj. } A| = |A|^2$$

$$A \rightarrow A^{-1}$$

$$\therefore |\text{adj } A^{-1}| = |A^{-1}|^2 = \frac{1}{|A|^2} = 4. \text{ Ans.}$$

Q.12 (A)

$$|\alpha - \beta| = \sqrt{10} \Rightarrow (\alpha + \beta)^2 - 4\alpha\beta = 10$$

$$\Rightarrow p^2 - 3p - 10 = 0 \Rightarrow (p-5)(p+2) = 0 \Rightarrow p = -2, 5. \text{ Ans.}$$

Q.13 (B)

$$T_n = n \cdot (2n)^2 \Rightarrow S_{10} = \sum_{n=1}^{10} 4n^3 = 4 \left(\frac{10 \times 11}{2} \right)^2 = (110)^2 = 12100. \text{ Ans.}$$

Q.14 (B)

Q.15 (A)

Taking log on both sides of given equations,

$$\log 3 (\log 3 + \log x) = \log 4 (\log 4 + \log y) \quad \dots\dots(i)$$

$$\text{and } \log x (\log 4) = (\log y) \log 3 \quad \dots\dots(ii)$$

From (i)

$$(\log 4)^2 - (\log 3)^2 = \log 3 \log x - \log 4 \log y \quad \dots\dots(iii)$$

From (ii)

$$\frac{\log x}{\log 3} = \frac{\log y}{\log 4} = \lambda \text{ (say)}$$

From (iv) and (iii)

$$(\log 4)^2 - (\log 3)^2 = \{(\log 3)^2 - (\log 4)^2\} \lambda$$

$$\therefore \lambda = -1$$

$$\therefore \log x = -\log 3 = \log \frac{1}{3} \Rightarrow \log y = -\log 4 = \log \frac{1}{4}$$

$$\therefore x = \frac{1}{3}, y = \frac{1}{4} \quad \text{Ans.}$$

Q.16 (ABCD)

Let $f(x) = ax^2 + bx + c$

$$\int_{\alpha}^{\beta} f(x) dx > 0, f(x) > 0 \Rightarrow f(x) = 0 \text{ has no real roots}$$

$$g(0) = f''(0) f(0)$$

$$\Rightarrow 12 = 2ac \Rightarrow ac = 6 \text{ also } a \text{ and } c \text{ are positive integers so possible values}$$

$$a = 6, c = 1, a = 1, c = 6; a = 2, c = 3; a = 3, c = 2$$

$$\text{Again } b^2 < 4ac \Rightarrow b^2 < 24 \text{ so } b \text{ can be } 1, 2, 3, 4$$

$$\text{Hence possible equations } = 4.4 = 16$$

$$\text{Also } f(x) > 0 \forall x \in \mathbb{R}$$

$$f(1) = a + b + c \quad \text{minimum } f(1) = 2 + 3 + 1 = 6$$

$$f(1) = a + b + c \quad f(1) = 2 + 3 + 1 = 6$$

$$\text{Maximum } f(1) = 6 + 4 + 1 = 11$$

Q.17 (ABCD)

$$\log_{x^2} 16 + \log_{2x} 64 = 6$$

$$\text{Equation is defined when } x > 0 \text{ and } x \neq \frac{1}{2}, 1$$

$$\text{Now, } \frac{1}{2} \log_x 4^2 + \log_{2x} 4^3 = 3 \Rightarrow \log_x 4 + 3(\log_{2x} 4) = 3 \Rightarrow 2 \log_x 2 + 6 \log_{2x} 2 = 3$$

$$\Rightarrow \frac{2}{(\log_2 x)} + \frac{6}{(\log_2 2x)} = 3 \Rightarrow \frac{2}{t} + \frac{6}{(1+t)} = 3 \quad \{\text{where } t = \log_2 x\}$$

$$\Rightarrow 2 + 2t + 6t = 3t + 3t^2 \Rightarrow 3t^2 - 5t - 2 = 0 \Rightarrow 3t^2 - 6t + t - 2 = 0$$

$$\Rightarrow 3t(t-2) + 1(t-2) = 0 \Rightarrow t = 2, -\frac{1}{3} \Rightarrow \log_2 x = 2, -\frac{1}{3}$$

$$\Rightarrow x = 4, 2^{-\frac{1}{3}}$$

Q.18 (A) P, T; (B) Q; (C) P; (D) Q

(A) $\alpha + \beta = \frac{a}{2}$ and $\alpha\beta = 2$

$$\sum_{n=1}^{\infty} \left(\frac{1}{\alpha} + \beta \right)^n = 3 \Rightarrow \left(\frac{3}{\alpha} \right)^1 + \left(\frac{3}{\alpha} \right)^2 + \left(\frac{3}{\alpha} \right)^3 + \dots = 3$$

$$\Rightarrow \frac{\frac{3}{\alpha}}{1 - \frac{3}{\alpha}} = 3 \Rightarrow 3 = 3\alpha - 9$$

$$\Rightarrow \alpha = 4 \Rightarrow \beta = \frac{1}{2}$$

$$\therefore \frac{a}{2} = \frac{9}{2} \Rightarrow a = 9$$

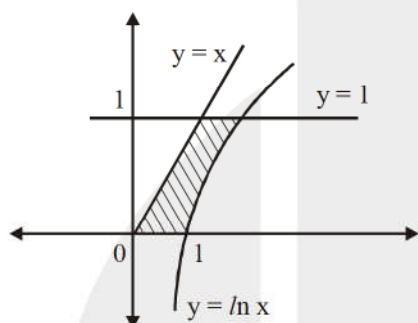
- (B) Let $f(x)=0$ has integral roots γ and δ
 $\therefore f(2)=15 \Rightarrow 3(2-\gamma)(2-\delta)=15 \Rightarrow (2-\gamma)(2-\delta)=5$
 $\therefore 2-\gamma=1, 2-\delta=5$
 $\Rightarrow \gamma=1, \delta=-3$ or $2-\gamma=-1, 2-\delta=-5$
 $\gamma=3, \delta=7$ (reject otherwise $\sin \alpha > 1$)

$$\therefore \gamma + \delta = \frac{-7 \sin \alpha}{3} \Rightarrow \frac{-7 \sin \alpha}{3} = -2 \Rightarrow \sin \alpha = \frac{6}{7} \Rightarrow \text{two values of } \alpha$$

$$\gamma \delta = \frac{\tan \beta}{3} \Rightarrow \frac{\tan \beta}{3} = -3 \Rightarrow \tan \beta = -9 \Rightarrow \text{two values of } \beta$$

$\therefore$ Total 4 ordered pairs of (α, β)

(C)



$$A = \left| \int_0^1 e^y dy - \frac{1}{2}(1.1) \right| = \left| e - \frac{3}{2} \right|$$

(D) $\frac{\Delta}{S-a}=3, \frac{\Delta}{S-b}=6, \frac{\Delta}{S-c}=2 \Rightarrow \frac{S-a}{\Delta} + \frac{S-b}{\Delta} + \frac{S-c}{\Delta} = \frac{1}{3} + \frac{1}{6} + \frac{1}{2} = 1 \Rightarrow S = \Delta$ and

$$\text{Now } \frac{S-a}{\Delta} \cdot \frac{S-b}{\Delta} \cdot \frac{S-c}{\Delta} = \frac{1}{36} \Rightarrow \frac{\Delta^2}{S\Delta^3} = \frac{1}{36} \Rightarrow \Delta = 6 = S$$

$$\Rightarrow \frac{6}{6-a} = 3 \Rightarrow a = 4$$

Q.19 [0004]

$$p = \frac{a+b}{2a-b} + \frac{c+b}{2c-b} = \frac{a + \frac{2ac}{a+c}}{2a - \frac{2ac}{a+c}} + \frac{c + \frac{2ac}{a+c}}{2c - \frac{2ac}{a+c}} = \frac{a+3c}{2a} + \frac{3a+c}{2c} = 1 + \frac{3}{2} \left(\frac{c}{a} + \frac{a}{c} \right) > 4$$

$$\text{Also, } \sqrt{\lambda \sqrt{\lambda \sqrt{\lambda \dots \infty}}} = \lambda^{\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots} = \lambda$$

$$\therefore \lambda = 4 \quad \text{Ans.}$$

Q.20 [0000]

Put $t = (5 + 2\sqrt{6})^{x^2-3}$

$$(5 - 2\sqrt{6})^{x^2-3} = \frac{1}{(5 + 2\sqrt{6})^{x^2-3}} = \frac{1}{t}$$

$$\Rightarrow t + \frac{1}{t} = 10 \Rightarrow t = 5 \pm 2\sqrt{6}$$

$$(5 + 2\sqrt{6})^{x^2-3} = 5 + 2\sqrt{6} \quad \text{or} \quad (5 + 2\sqrt{6})^{x^2-3} = 5 - 2\sqrt{6}$$

$$x^2 - 3 = 1 \quad \text{or} \quad x^2 - 3 = -1$$

$$x^2 = 4 \quad \text{or} \quad x^2 = 2$$

$$x = \pm 2 \quad \text{or} \quad x = \pm \sqrt{2}$$

Sum of solutions = 0 **Ans.**

Q.21 [0004]

Sol. $2x^{\log_4 3} + 3^{\log_4 x} = 27 \Rightarrow \log_4 x = 2$

$$\sqrt{x} = 4 \quad \text{Ans.}$$

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